Public Issue Resolution Tracker

Report a civic issue, watch it get fixed in public.

TEAM NAME: TINKERERS

TEAM ID: TH22-0072

TEAM MEMBERS

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PS ID: WEB03- Public Issue Resolution Tracker

The problem affects real people

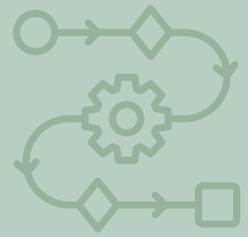
WHO FACES IT?

- **Primary user group:** Daily commuters, local residents, and municipal field officers
- The problem occurs in connecting public complaints to authorities when civic amenities fail
- eg: Guwahati's recurring waterlogging and drainage failure

WHY DOES IT MATTER?

- Complaints pile up with no visibility into who is handling them or what stage they are at
- Cost, delay or risk caused:Duplicate issues wastes time and genuine issues get delayed.

Our project groups duplicate complaints, track progress, and verify fixes



SOLUTION

A public portal where citizens submit issues, likely duplicates are linked instead of lost and every report is tracked from submission to resolution.

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FEATURES

- Civic Issue Reporting
- Role-based Access Control
- Public Feed with Map & Filters
- Issue Lifecycle Tracking
- Real-time Duplicate Detection
- Officer Dashboard
- Email verification and Updates at Every Resolution Stage
- AI Powered Resolution Verification

https://www.loksamadhan.online

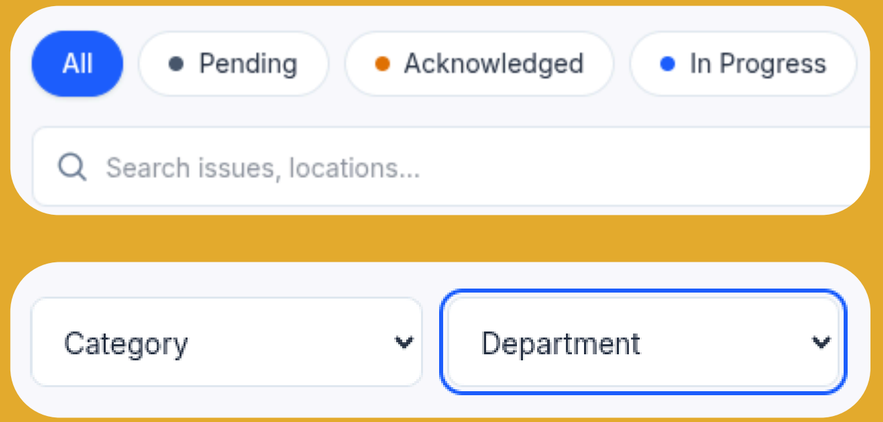
TECHNICAL IMPLEMENTATIONS

- **Frontend:**
  - Core Framework: - React 19
  - Build Tool:- Vite 8
  - Styling & Design System: - Tailwind CSS
  - Maps & Geo-Visualization:- Leaflet
- **Backend:**
  - Runtime: Node.js
  - Web Framework:- Express.js
  - Authentication & Security- JWT, CORS, bcryptjs, OAuth
  - File Upload Processing: Multer
- **Database:** MongoDB
- **Main data flow:** Citizen reports → Issue stored with location → Officer assigns & updates → Public tracks status.
- **External API:**
  - OpenStreetMap: Provides the interactive map tiles displayed
  - Cloudinary: Stores uploaded issue photos
  - MyMemory & Google Cloud Translation API: Traslation
  - Groq Cloud API: AI Vision & LLM
  - Google OAuth 2.0 & Identity APIs: Authentication
  - SMTP Email Service: Notifications & OTP Verification
  - WhatsApp Click-to-Chat URL API: share civic issues directly to WhatsApp

CHALLENGE CARD

USEFUL FILTERS

The public issues are filtered as per status, location, category and department in the feed.



Today’s Progress, Tomorrow’s Plan

Designed to serve both citizens and public authorities

- **Enforced Public-Safe Data Serialization:** Citizen personal information is never included in public API responses.
- **Resolution requires a note or evidence:** An officer cannot mark an issue as Resolved without documenting how it was resolved.

LIMITATIONS

- No integration with actual municipal ERPs, GIS databases, or government grievance redressal portals
- No phone/Aadhaar identity verification for citizens

THANK YOU